DEEPAK KATUKURI

AI ENGINEER · AGENTIC AI · RAG



CONTACT

✉ deepakkatukuri@gmail.com
☎ +49 1629741795
◆ Potsdam (Berlin area), Germany
◉ github.com/Deepak589
◉ linkedin.com/in/deepak-katukuri

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

LLM & RAG: Retrieval-Augmented Generation, LangChain, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, pgvector, CLIP, DistilBERT, sentence-transformers, contrastive learning, embeddings, Ollama

Deep Learning: PyTorch, TensorFlow, CNNs, fine-tuning, transformers, Hugging Face, timm, EfficientNet

Cloud & DevOps: AWS (EC2, RDS, VPC, Elastic IP), Kubernetes (K3s, HPA, Ingress, Secrets), Docker, Docker Compose, GitHub Actions, CI/CD, Vercel, Render

Full-Stack: FastAPI, REST APIs, Flask, React, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, Recharts

ML & Data: Scikit-Learn, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn

Tools: Git/GitHub, pytest, Jupyter, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

AI Engineer and Data Science MSc candidate, with 2.5 years of production software engineering behind the ML work. Built a hand-rolled hybrid RAG retriever, then validated and scaled it on LangChain against a 55-question golden set. Ships a live full-stack digital health app (React, FastAPI, PostgreSQL) in production, and trained a CLIP dual-encoder from scratch for multimodal retrieval. Looking for an English-speaking working student (Werkstudent) AI / Agentic Engineering role in Germany.

PROJECTS

RAG Pipeline: Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama · github.com/Deepak589/RAG_pipeline

- Built a hand-rolled hybrid retriever from scratch over 4 iterative stages: BM25 plus a dense bi-encoder, RRF fusion, cross-encoder reranking. Reached 0.833 recall@5 and 0.672 MRR on a 55-question labeled golden set before adopting any framework.
- Rebuilt the same retriever on LangChain using the identical corpus and labels, with the framework as the only changed variable. Caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer and fixed it to 0.854 recall@5, so both implementations matched before scaling up.
- Scaled the corpus ~100x, from 7 to ~740 documents and ~3.5GB across arXiv, SEC filings, Wikipedia and scanned PDFs, using a resumable ingestion pipeline that routes by file type into a pgvector-backed dense store.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · FastAPI · PostgreSQL · Docker · live at ra-project-indol.vercel.app

- Deployed a live full-stack health app as two tiers: a React/Vite/Tailwind frontend on Vercel, and a FastAPI + PostgreSQL backend (4 logical schemas) running as a Docker container on Render, with JWT auth and token refresh.
- Containerised the local stack with Docker Compose: the API, PostgreSQL 16 gated by a pg_isready healthcheck, and an optional pgAdmin profile. Alembic migrations run on container start, so the whole environment comes up with one command.
- Built a rule-based recommendation engine on top of USDA FoodData Central, plus a weekly Recharts dashboard that turns raw meal and symptom logs into specific suggestions for the user.

Food Image-to-Recipe Retrieval: Multimodal Deep Learning

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a dual-encoder retrieval model that maps food images (frozen CLIP ViT-B/32) and recipe text (DistilBERT, with separate ingredient and instruction streams) into a shared 1024-d space, trained with symmetric InfoNCE contrastive loss.
- Reached a validation median rank of 10/200 with semantically correct top-5 results on a 13.5K-image Kaggle dataset, and cut training time ~10x by pre-computing and caching CLIP image features.

CloudNotes: Kubernetes & AWS Deployment Platform

AWS EC2 + RDS · K3s · GitHub Actions · Traefik · MSc cloud computing, team of 3 · github.com/Deepak589/cloudnotes

- Owned the Kubernetes layer on a 3-person team. Wrote the Deployment with CPU and memory requests and limits, a ClusterIP Service, a Traefik Ingress, readiness and liveness probes, and an HPA that scales 2 to 5 replicas at 60% CPU. Credentials come from Kubernetes Secrets, never from the image.
- Built the GitHub Actions pipeline that replaced the team's manual deploys: Docker buildx multi-architecture build (linux/amd64 and linux/arm64), push to Docker Hub, SSH to EC2, then kubectl set image. Every release is gated on kubectl rollout status, so rollouts finish with no downtime.
- Provisioned the AWS environment: EC2 t3.medium, Elastic IP, VPC, security groups and RDS MySQL. Costed managed Kubernetes against K3s and picked K3s for a single-node workload at about $86/month.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty. The OneShield object copy from the stored object to the loading object failed silently when another bond shared that bond type under a different bond ID. Kept records current with a weekly backfill script while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern
Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.